JAWAHARLAL COLLEGE OF ENGINEERING & TECHNOLOGY
JAWAHAR GARDENS, LAKKIDI MANGALAM, PALAKKAD
(An ISO 9001:2015 Certified Institution)

DEPARTMENT OF COMPUTER SCIENCE (CYBERSECURITY)

AI-BASED CVE VULNERABILITY ANALYZER

ABSTRACT

This project proposes the development of an AI-based cybersecurity system that is trained using recent Common Vulnerabilities and Exposures (CVEs) and other vulnerability-related data. With new vulnerabilities being discovered every day, traditional security methods may not be able to analyze and prioritize threats quickly. The proposed system collects vulnerability information from reliable sources, including CVE descriptions, severity scores, affected software and hardware, vulnerability types, attack methods, and available remediation techniques. The collected data is cleaned, processed, and used to train an AI model using machine learning and natural language processing techniques. The model will analyze newly reported vulnerabilities, classify their severity, identify possible security impacts, and recommend appropriate mitigation measures. The system can be regularly updated with newly published CVEs, allowing the model to learn from recent vulnerability trends and improve its ability to recognize emerging threats. It can also help security teams prioritize critical vulnerabilities based on their potential risk and impact. The main goal of this project is to reduce the time and manual effort required for vulnerability analysis while providing up-to-date and intelligent security insights. Overall, the proposed system demonstrates how AI and continuously updated vulnerability data can be combined to improve vulnerability detection, risk assessment, prioritization, and cybersecurity response.

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